

Circular 3D friction under slow and rapid container motion

The project now has a circular, coupled Coulomb contact solver with explicit residual-driven iteration work, prescribed wall reversals and full 3D inertia. The frozen study qualifies **three of six** trajectory references. It does not qualify fast shaking with 27 spheres or either random-hull case.

The native off-centre test found and fixed the initial world-inertia frame: `setCenterOfMassTransform` initializes the rotated inverse inertia before the first collision. A separate eight-body packed test verifies that a prescribed 100 to -100 m/s reversal reaches contacts on the first update after its command. Seven 3D friction regression tests check diagonal sliding, sticking, slow and rapid transition to rolling, off-centre full-tensor contact, 27-body shaking and rejection on an exhausted contact budget. Eight native analytic friction cases check circular sliding and retained normal/tangent coupling.

Contact equations and scope

For each contact, the impulse has one normal and two tangential components. Let K be the assembled contact mobility, retaining all inter-body, lever-arm and normal/tangent coupling. The frozen velocity problem is

$$w = Kp - b, \quad p_n \geq 0, \quad w_n \geq 0, \quad p_n w_n = 0.$$

Sticking and sliding use a circular friction limit:

$$\|p_t\| \leq \mu p_n, \quad \begin{cases} w_t = 0, & \|p_t\| < \mu p_n, \\ p_t = -\mu p_n w_t / \|w_t\|, & \|w_t\| > 0. \end{cases}$$

Normal complementarity is separate from the friction disk. An associated cone energy QP would alter the normal law and is not substituted here. One coefficient covers sticking and sliding. The current lane is inelastic (normal restitution zero) and has no tangential spring or independent angular contact couple. Those energy-storing channels are tested separately in the elastic-contact study.

Block iterations propagate through sparse columns of K ; exact 2D disk subproblems and semismooth Newton acceleration use the same equations. Dense Bullet assembly is still quadratic; Newton acceleration is limited to at most 512 rows. Every eight sweeps, a velocity-scaled projection residual is checked against 1e-8 m/s. The 4096-sweep argument is a maximum budget. Unconverged runs reject instead of falling back to the upstream friction pyramid. Split position correction solves normal equations only and applies no tangential correction.

The declared contact geometry tolerance is 1e-9 m. Positive gaps within this tolerance are treated as touching; tiny penetrations within it are not projected. This prevents conflicting gap/time targets at nearly redundant face points. It is a numerical geometry tolerance, not material elasticity. Relative linear and angular tip speeds adjust the native travel-guard timestep; its rule is a conservative heuristic, not a proof of swept collision detection for every shape.

Frozen results

All scenes last 0.12 s, sampled every 0.01 s. Synthetic pair friction is 0.4, normal restitution zero and gravity -9.81 m/s². Wall speeds are 0.2 or 20 m/s; shaking reverses at 0.04 and 0.08 s. Random rotating hulls additionally use 10 rad/s wall rotation. The archived plan and execution source were committed before execution. There are 76 retained attempts: 52 complete histories and 24 explicit native rejections. None is discarded.

Scene	Reference gate	Selected effort	Native median seconds
slow shake27 spheres	Pass	coarse	0.005612
fast translate27 spheres	Pass	coarse	0.234430
fast shake27 spheres	Fail	None	—
fast shake8 boxes	Pass	medium	0.375185
fast shake8 hulls42	Fail	None	—
fast rotate shake27 hulls7301	Fail	None	—

Scene	Reference gate	Selected effort	Native median seconds
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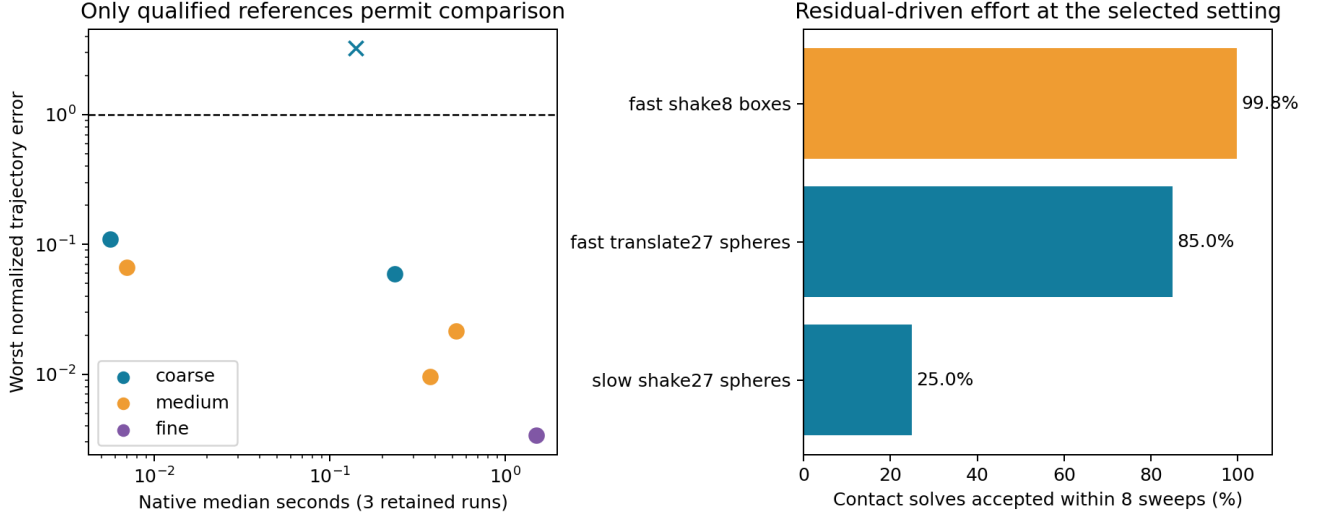


Figure 1: Trajectory budgets and effort

The trajectory budgets are 5 mm position RMS, 0.05 m/s velocity RMS, 0.1 rad/s angular-velocity RMS and 0.01 rad quaternion geodesic RMS. Both adjacent edges of the three finest declared refinement levels must meet **one quarter** of each budget. The complete coarser ladder remains visible; it is not a requirement that every coarse level qualify. No failed refinement level is skipped. Native physical gates require unit quaternion error at most $1e-12$, container surface excess at most 2 mm at every internal update and total energy change minus boundary work at most 1 J.

At qualified references, candidates use travel fractions 0.15, 0.06 and 0.015 with the same material/contact equations. Three retained repetitions give each candidate’s median; repeated states are required to be bitwise identical. The least-cost candidate that passes every gate is selected. No recommendation is made at an unqualified reference. This selection is offline; the native iteration and travel guard adapt during integration, but no certified local-error controller or cache-preserving rollback is claimed.

Fast 27-sphere shaking passes frozen contact residuals yet fails trajectory refinement: its last angular RMS difference is about 0.499 rad/s and velocity RMS difference about 0.079 m/s. This is direct evidence that a successful frozen contact solve and containment are insufficient to certify a trajectory. Both random-hull cases reject all twelve attempts each.

A subsequent independent review identified an upstream RHS inconsistency: normal equations and body writeback include a gyroscopic angular-velocity increment, while upstream tangent equations omit it. The adapter correction and zero-free-slip two-body regression are a **separate subsequent change**. This archive retains the original 24 hull rejections; it is not retrospectively fixed. Its accepted spherical cases have zero anisotropic gyroscopic increment; box spin is negligible. The matrix energy check in this frozen revision describes the assembled contact problem; global energy/work is also checked independently.

All timing runs used the same workspace while other research ran concurrently. These are descriptive costs, not a universal speedup, controlled hardware ranking or Vektor compiler benchmark. All parameters are synthetic; numerical agreement and fitting do not authenticate rubber or other materials.

Reproduce and audit

Execution source: 7c279768f6518e66b1ee620d44bd9b37309b1b5f. Pinned unmodified upstream: Bullet 3.25 Float64, 2c204c49e56ed15ec5fcfa71d199ab6d6570b3f5. `results/execution-source.zip` contains the exact code/plan with SHA-256 receipts; `results/scenes.json` contains full geometry; `results/traces.zip` retains

every completed history and rejection. `results/summary.json` records all diagnostics, refinement edges, candidate qualifications, timing and source/binary hashes.

Run `python -m research.audit_spatial_friction` to independently recompute physical gates, refinement, candidate selection and timing receipts. Historical 3D normal gains and the earlier failed 102-history friction study are unchanged. The Vektor integration target remains `vektor-flow/bootstrap`, paired with spec; this public Python/C++ evidence is not a completed Vektor port.